



- 2 A particle P of mass m kg moves along a horizontal straight line against a resistive force of magnitude $2mv^3$ N, where v ms⁻¹ is the velocity of P at time t s. When $t = 0$, $v = 1$.

(a) Find an expression for v in terms of t .

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(b) Find the displacement of P from its initial position when $t = 6$.

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